

DengAI — Predicting Disease Spread

Metrics and Models

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Contents

Metrics	1
Models	3

Metrics

The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- Reported over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- Both cities are pooled into one number, so **San Juan dominates** — it has 63% of the test weeks *and* roughly four times the case counts

Why MAE and not RMSE

The target is extremely skewed:

	San Juan	Iquitos
mean cases/week	34.2	7.6
median	19	5
max	461	116
skewness	4.48	4.00

In both cities the **worst 10% of weeks hold 43% of all cases**.

What “skewed” actually means

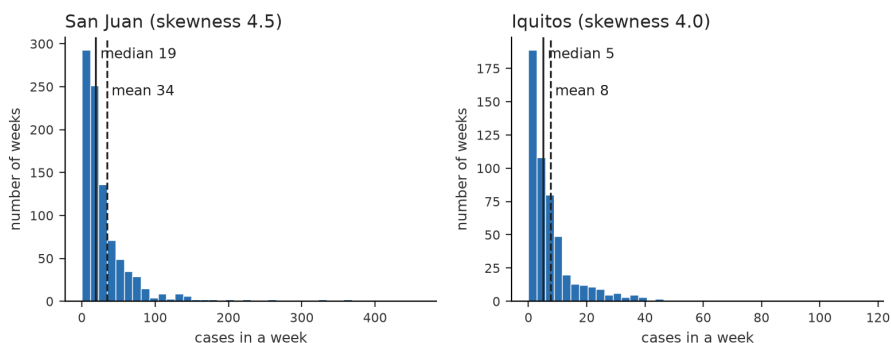


Figure 1: Weekly dengue cases: most weeks are quiet, a few outbreak weeks form a long tail

What “skewed” actually means (cont.)

Most weeks have few cases; a handful of outbreak weeks have hundreds. The distribution is lopsided, with a **long tail to the right**.

- **The mean gets pulled toward the tail.** San Juan’s mean is 34 cases, but a typical (median) week has 19. About **73% of weeks are below the mean**.
- **Skewness is a number for this lopsidedness.** 0 means symmetric, like a bell curve; above 1 already counts as strongly skewed. Ours is **4.5 and 4.0**.
- **Everyday analogy:** income. A few billionaires raise the *average* income, but the *median* still describes the typical person.

For us this means that the rare outbreak weeks would dominate any metric that punishes big errors heavily.

Why MAE and not RMSE (cont.)

RMSE squares the errors, so a single epidemic week with 461 cases contributes as much as **~500 ordinary weeks** being off by one.

- Under RMSE the model is dragged into fitting a handful of outbreaks
- MAE weights every week equally
- Statistically: **MAE is minimised by the conditional median, RMSE by the mean** — and for a skewed count distribution the median is the more robust target

This also explains a behaviour we see later: good models here are *conservative*.

Models

The pipeline has four branches

Different model families need different preprocessing, so `make_city_pipeline` builds one of four shapes:

Branch	Steps	Models
<code>tree</code>	impute → history → cyclical → select → model	CatBoost, RandomForest
<code>dense</code>	tree steps + impute + scale	Ridge, Lasso
<code>time_series</code>	impute → history → date + scaled weather	Prophet, SARIMAX
<code>seasonal</code>	calendar columns only	Seasonal median, Shape×level

Why `dense` exists

The `tree` branch leaves **NaNs** in the lag/rolling warm-up — the first 26 weeks have no 26-week history.

- Tree models (sklearn ≥ 1.4) **handle NaN natively**
- RidgeCV raises `ValueError: Input X contains NaN`

So linear models get two extra steps: `SimpleImputer(median)` then `StandardScaler`. Scaling matters for them too, since Ridge and Lasso penalise coefficients and are not scale-invariant.

Why `seasonal` exists

The tree branch **drops** `city`, `week_start_date` and `weekofyear` before the model.

But the two calendar baselines need exactly those columns. Rather than weaken the main branch, they get their own: a `ColumnTransformer` passing through `year` and `weekofyear`, nothing else.

These models **never see the weather at all** — which is the point: they are the baseline any weather-driven model should have to beat.

The nine models

Model	Family	Core assumption
CatBoost	gradient boosting	non-linear effects of lagged weather

Model	Family	Core assumption
XGBoost	gradient boosting	as CatBoost, on RFE-selected features (side experiment)
RandomForest	bagged trees	same, without boosting
Ridge	linear + L2	smooth linear response, all features shrunk
Lasso	linear + L1	linear, but most features are irrelevant
Prophet	additive time series	trend + annual seasonality + regressors
SARIMAX	ARIMA + exog	autocorrelated errors + Fourier seasonality
Seasonal median	calendar only	this week behaves like the same week in past years
Shape $\times$ level	calendar only	seasonal <i>shape</i> $\times$ typical annual <i>total</i>

Results — holdout MAE

Model	San Juan	Iquitos
CatBoost	15.02	6.27
RandomForest	15.16	8.39
XGBoost + RFE (side experiment)	17.21	3.54
Shape $\times$ level	20.72	3.39
Lasso	21.16	4.50
Prophet	21.78	7.16
Seasonal median	23.49	4.41
Ridge	24.46	5.48
SARIMAX	24.56	4.61

Different winners per city — so the final submission uses **CatBoost for San Juan**, **SARIMAX for Iquitos**.

Side experiment — XGBoost and feature selection

A separate notebook (`experiments/xgboost_feature_selection.ipynb`) asks whether all 127 engineered features help, or whether fewer would do: XGBoost tuned on chronological CV (3 folds of 52 weeks) with three feature-selection strategies.

Feature selection	San Juan	Iquitos
All 127 features	17.8	7.2
XGBoost importance, top 100	17.0	7.1
Ridge RFE	14.5 (50 features)	6.7 (10 features)

CV MAE. Recursive elimination with a *linear* model chose features better than the trees' own importances did.

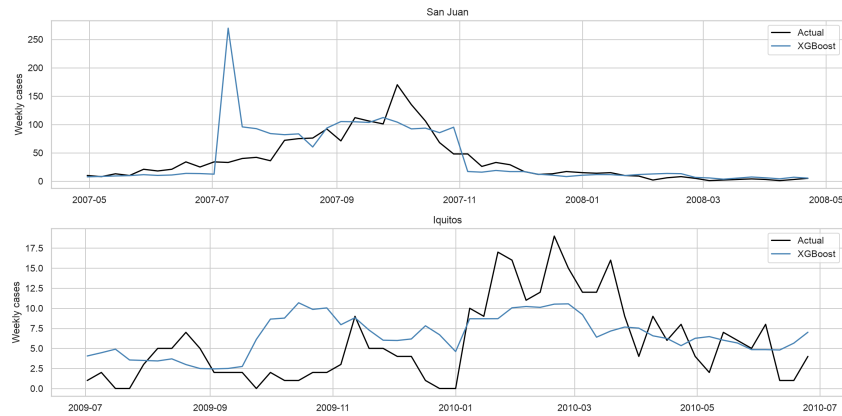


Figure 2: XGBoost (blue) against actual cases (black), 52-week holdout

XGBoost on the holdout year

Holdout MAE **17.21** in San Juan (third; CatBoost 15.02) and **3.54** in Iquitos: the **best weather-based model** there (SARIMAX 4.61, CatBoost 6.27), with only ten long-window features. It follows the 2007 outbreak but under-predicts both peaks.

What drives the CatBoost predictions?

Share of CatBoost feature importance, grouped by what the feature measures

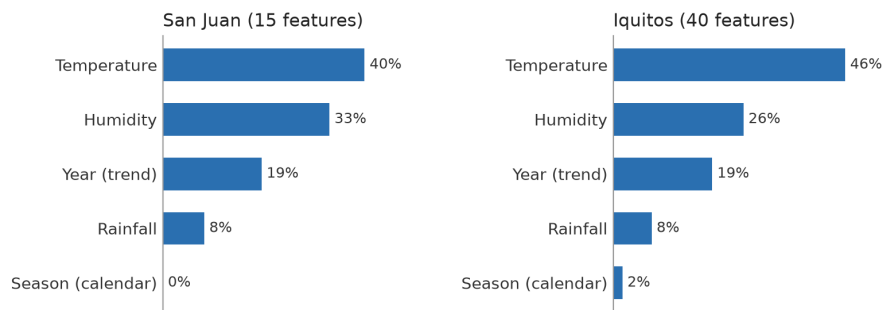


Figure 3: CatBoost feature importance, grouped by what each feature measures

Feature importance — what the model is telling us

- **Temperature and humidity carry about 70% of the importance** in both cities. Mosquitoes breed and bite more when it is warm and humid.
- **Every top feature is a long average over 8–26 weeks**, not this week's

weather. Weather acts with a delay: the mosquito population has to build up first.

- **Rainfall matters less than expected** (about 8%). Temperature and humidity already capture most of what the rain does.
- **year alone takes about 19%**. Case levels drift from year to year for reasons that are not weather.

Importance tells us what the model *uses*, not what *causes* dengue.

Why weather is not the whole story

Feature importance and the results table point the same way. In Iquitos the calendar-only **Shape × level** model (3.39) beats every weather model:

- Climate reliably predicts **when** cases rise within a year
- It explains very little about **how many**

Epidemic magnitude is driven by which dengue **serotype** is circulating and by population immunity — neither is in this dataset.

So a model that predicts the *shape* from the calendar and refuses to guess the *level* from weather is not naive. It is **honest about what the data supports**.

Summary

Metrics — MAE, because the target is skewed (skewness 4.5): a few outbreak weeks would dominate RMSE.

Models — nine models across four pipeline branches, because tree, linear, time-series and calendar families need different preprocessing. Winners differ by city. A side experiment with XGBoost confirms that feature selection matters: 50 features beat 127 in San Juan, and 10 are enough in Iquitos.

Feature importance — CatBoost relies mostly on temperature and humidity averaged over months. Weather tells us *when* cases rise, not *how many*.